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Quantifying the combined effects of pronase and benzalkonium chloride in removing late-stage *Listeria monocytogenes*–*Escherichia coli* dual-species biofilms

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ABSTRACT

This work presents the assessment of the effectivity of a pronase (PRN)-benzalkonium chloride (BAC) sequential treatment in removing *Listeria monocytogenes*–*Escherichia coli* dual-species biofilms grown on stainless steel (SS) using fluorescence microscopy and plate count assays. The effects of PRN-BAC on the occupied area (OA) by undamaged cells in 168 h dual-species samples were determined using a first-order factorial design. Empirical equations significantly ($r^2 = 0.927$) described a negative individual effect of BAC and a negative interactive effect of PRN-BAC achieving OA reductions up to 46%. After treatment, high numbers of remaining attached and released viable and cultivable *E. coli* cells were detected in PRN-BAC combinations when low BAC concentrations were used. Therefore, at appropriate BAC doses, in addition to biofilm removal, sequential application of PRN and BAC represents an appealing strategy for pathogen control on SS surfaces while hindering the dispersion of live cells into the environment.

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fluorescence microscopy;
Listeria monocytogenes;
Escherichia coli; pronase

Introduction

Biofilms are considered the main structure in which bacteria are usually found in the environment (Costerton et al. 1995; Vlamakis 2011; Winkelströter et al. 2013; Abdallah et al. 2014). These structures can be defined as sessile communities of microorganisms residing in a self-secreted matrix (Costerton et al. 1995). Bacteria in this state are able to develop resistance to environmental insults (Davey and O'toole 2000; O'Toole et al. 2000). In the food industry such a phenomenon could lead to a higher tolerance to disinfectants (Bridier et al. 2011), causing eventual foodstuff cross-contamination (Bridier et al. 2015; Muhterem-Uyar et al. 2015).

Listeria monocytogenes is a Gram-positive, environmentally ubiquitous bacterium commonly living in nature in soils rich in plant-decay matter. In humans it can cause listeriosis, a disease with an increasing incidence tendency in Europe over the last years (EFSA 2015). The invasive form of this illness can severely affect newborns, the elderly, pregnant women and individuals with a compromised immune system, with symptoms that may vary from septicaemia, neurological harm and miscarriage (Freitag et al. 2009). This pathogen can be also found adhered to

surfaces of sanitary and food industrial settings (Gandhi and Chikindas 2007; Rodríguez-López et al. 2015; Zhang et al. 2016). In the latter case, it can persist for long periods of time, thus becoming contamination hotspots for food products (Carpentier and Cerf 2011). Besides, this microorganism can easily associate with other bacterial species forming part of complex microbial communities (Carpentier and Chassaing 2004; Rodríguez-López et al. 2015), which can confer higher resistance to biocides (Kostaki et al. 2012; Giaouris et al. 2013).

For biofilm control, chemical-based treatments using peroxides, electrolysed water, organic acids and quaternary ammonium compounds (QACs) have been used alone (McCarthy and Burkhardt 2012; da Silva and De Martinis 2013) or combined (Vázquez-Sánchez et al. 2014) to prevent both the formation and removal of already formed structures. Among QACs, benzalkonium chloride (BAC) is usually preferred due to its low cost and its bactericidal effects which affect the permeability of the cell membrane and induce irreversible cell damage (McDonnell and Russell 1999; Tezel and Pavlostathis 2015). Nonetheless, in *L. monocytogenes* it has been proved that biofilms present higher resistance to BAC in comparison to the planktonic

counterparts (Ibusquiza et al. 2011). This fact has also been observed in Gram-negative species (Houari and Di Martino 2007; Giaouris et al. 2013).

Enzymatic treatments have been proposed as an efficient, environmentally friendly possibility for surface cleaning as well as a biofilm-preventive strategy in industrial settings (Lequette et al. 2010) or for decontamination of medical devices to prevent infection *via* indwelling devices (Marion et al. 2006; Stiefel et al. 2016). Since enzymes specifically cleave biological molecules such as proteins, polysaccharides and DNA, they are used to degrade components of the biofilm matrix affecting the stability of the structure and therefore producing eventual biofilm dispersion (Kaplan 2014). Nevertheless, enzymes used for anti-biofilm procedures in the food industry generally lack biocidal activity, making them inappropriate for bactericidal purposes (Meireles et al. 2016). To overcome this problem, a combination of enzymatic and chemical approaches would be desirable, since the action of the enzyme would positively contribute to the killing capacity of the disinfectant (Meireles et al. 2016). In this regard, the study by Kaplan (2009) demonstrated that DNase I sensitised 24 h *Staphylococcus aureus* biofilms to antiseptics and disinfectants. In addition, a recent study demonstrated that the combination of pronase (PRN) or DNase I with BAC was able to remove 48 h *L. monocytogenes*–*E. coli* biofilms (Rodríguez-López et al. 2017).

For biofilm quantification after disinfection, most of the methods are based on the remaining adhered cells recovered from samples followed by cultivable cell counting on agar plates. However, these agar plating-based techniques present several limitations (Sutton 2011) and they do not provide information regarding morphological features. Therefore, it is usual to incorporate microscopic assays, mainly epifluorescence and confocal laser scanning microscopy (CLSM) to describe 2-D and 3-D biofilm structures, respectively.

In epifluorescence microscopy, an issue of concern is how to select the appropriate 2-D parameters to give an accurate description of the effects, being easily related to biological processes to obtain meaningful conclusions. Several authors have concluded that the areal porosity is a good 2-D structural outcome, being intuitive, easy to interpret and with clear biological meaning (Christensen et al. 2001; Jackson et al. 2001; Beyenal et al. 2004; Dusane et al. 2008). Beyenal et al. (2004) suggested parameter selection according to the specific process under study. Following this criterion, the biological meaning of areal porosity, and by extension the occupied area, seems to be especially appropriate to describe biofilm removal.

This work aimed to study the effectiveness of combining pronase (PRN) and BAC for the removal of late-stage *L. monocytogenes*–*E. coli* dual-species biofilms grown on

stainless steel (SS), mimicking real industrial conditions where biofilms are formed after long periods. PRN was selected since it has been reported that the biofilm matrix of *L. monocytogenes* is mainly constituted of proteins (Longhi et al. 2008; Nguyen and Burrows 2014) and PRN is efficient at removing early-stage *L. monocytogenes*–*E. coli* dual-species biofilms (Rodríguez-López et al. 2017). Enzyme solutions were applied at room temperature to further simulate realistic environmental conditions. The assessment of the effects was performed by combining microscopy and image analysis with classical microbiology methods. To ascertain the feasibility of the microscopy approach, the method was firstly statistically evaluated in two different dual-species biofilms: *L. monocytogenes*–*E. coli* and *L. monocytogenes*–*Pseudomonas fluorescens*.

Materials and methods

Bacterial strains

Two different consortia were used. The first was formed by *L. monocytogenes* A1–*E. coli* A14, both isolated from a fish processing plant in a previous survey (Rodríguez-López et al. 2015). The second was formed by a strain of *L. monocytogenes* G1, isolated from a cheese processing plant, kindly provided by Dr Luisa Brito (Leite et al. 2006) and *P. fluorescens* B52, one of the species commonly isolated in the dairy industry, kindly provided by Dr Carmen San José (Allison et al. 1998). These consortia were chosen based on their relevance in the fish and dairy industries, and their capability to form dual-species biofilms. From now on consortia used will be referred to as fish industry and dairy industry consortia for A1–A14 and G1–B52 biofilms, respectively.

In all cases, stock cultures were maintained at -80°C in brain-heart infusion broth (BHI; Biolife, Milan, Italy) containing 50% glycerol 1:1 (v v⁻¹) mixed. Laboratory stocks were kept at -20°C in trypticase soy broth (TSB; Cultimed, Barcelona, Spain) containing 50% glycerol 1:1 (v v⁻¹) mixed.

Set-up of dual-species biofilms

One hundred μl of laboratory stocks were cultured overnight in 5 ml of sterile TSB at 37°C for *L. monocytogenes* and *E. coli* and at 25°C for *P. fluorescens* and subcultured once so as to ensure proper activation.

Inocula preparation was performed as follows. Briefly, the Abs₇₀₀ of cultures was adjusted to 0.1 ± 0.001 with sterile phosphate buffered saline (PBS), corresponding to a bacterial concentration of $\sim 10^8$ CFU ml⁻¹ according to previous calibrations. Adjusted cultures were further diluted in sterile mTSB (TSB supplemented with 2.5 g l⁻¹ of glucose (Vorquímica, S.L., Vigo, Spain) and 0.6%

yeast extract (Cultimed)) until a final concentration of $\sim 10^4$ CFU ml⁻¹ was obtained and 1:1 (v v⁻¹) mixed.

Biofilms were grown on 10 × 10 × 1 mm AISI 316 stainless steel (SS) coupons (Comeisa, Vigo, Spain). Pre-treatment of coupons included individual washing with industrial soap (Sutter Wash, Sutter Ibérica, S.A., Madrid, Spain) to remove grease residues, thoroughly rinsing with tap water with a final rise with deionised water and sterilised by autoclaving them at 121°C for 20 min. Coupons were individually placed into a flat-bottomed 24-well plate and each well was inoculated with 1 ml of the corresponding culture. Plates were incubated in a humidified atmosphere at 25°C statically for 2 h for initial adhesion, and then in constant shaking at 100 rpm.

In all situations, coupons were aseptically collected and briefly immersed in 1 ml of sterile PBS to remove loosely attached cells before any assay was performed.

Plate count assays

In all cases, *Listeria*-PALCAM (Liofilchem, Roseto degli Abruzzi, Italy) was used to select *L. monocytogenes*; chromogenic *Escherichia coli* agar (Cultimed, Barcelona, Spain) supplemented with 5 mg l⁻¹ of vancomycin and cefsulodine (Sigma-Aldrich, St Louis, MO, USA) to isolate *E. coli*; and a *Pseudomonas* agar base (PAB) with a CFC supplement (Liofilchem) for *P. fluorescens*. Plates were incubated for 24–48 h at 37°C for chromogenic agar and PALCAM whereas 25°C was preferred for PAB.

Attached viable cultivable cells (AVC) were harvested from coupons by scraping using two sterile cotton swabs moistened in sterile buffered peptone water (BPW; Cultimed, Barcelona, Spain). Swabs were then suspended in 2 ml of BPW and vortexed vigorously for 1 min in order to release cells, serially diluted in BPW and spread onto agar plates for AVC determination. In reproducibility, repeatability and biofilm formation kinetic assays AVC values were expressed in CFU cm⁻² whereas in the enzyme-disinfectant assays they were expressed in log CFU cm⁻².

Viable cultivable cells (RVC) released into the neutralising solution (preparation detailed later) were determined after treatment by performing direct serial dilution of the solution in BPW and spreading onto appropriate solid media. Outcomes were expressed in log CFU ml⁻¹.

Fluorescence microscopy and image analysis assays

Samples (SS coupons) were stained for 15 min with FilmTracer™ LIVE/DEAD® Biofilm Viability Kit (Life Technologies, Eugene, OR, USA). The staining solution contained 0.75 µl of Syto9 and 0.25 µl of propidium iodide in 1 ml of filter sterilised deionised water. Fifty microlitres of this solution were used for each staining each coupon and allowed to remain for 15 min in the dark. After that, coupons were washed three times in 1 ml of sterile MilliQ water. Coupons were then air dried and visualised in a Leica 6000DM (Leica, Wetzlar, Germany) epifluorescence microscope using 10× ocular lenses and a 40× objective.

From each sample, a randomly chosen field was considered as the starting point to automatically acquire images using a Leica DFC365 FX camera. Each image set was composed of three mosaics of 25 12-bit images covering a total surface of 1.92×10^6 µm². Image analysis was then performed using the Integrated Morphometry Analysis module of the Metamorph MMAF software (Molecular Devices, Sunnyvale, CA, USA) in order to determine the occupied area (OA) by undamaged (green) cells.

The results of image analysis in biofilm formation were expressed as the percentage occupied area (POA) of the mosaic whilst in repeatability, reproducibility and enzyme-disinfectant experiments OA outcomes were expressed in mm².

Dual-species biofilm formation kinetics

Samples of both consortia were collected after incubation for 24, 48, 72, 96 and 168 h. At each sampling time, three coupons were used for plate counts and three more for microscope analysis as described above.

Effects of sequential pronase (PRN) – benzalkonium chloride (BAC) treatments on 168 h A1–A14 biofilms

PRN, BAC and neutralising solution preparation

Pronase (PRN; from *Streptomyces griseus*, Roche, Mannheim, Germany) was prepared at the concentrations listed in Table 1 using 0.1 M Tris-HCl (Sigma Aldrich) buffer at pH = 7.5 ± 0.2 and then filter sterilised through a 0.2 µm pore diameter syringe filter (Sartorius, Göttingen, Germany). Solutions were kept at –20°C until use. Benzalkonium chloride (BAC; Guinama, Alboraya, Spain) was prepared at the concentrations also listed in

Table 1. Pronase (PRN) and benzalkonium chloride (BAC) concentrations and their corresponding encoded values used in the factorial experimental approach.

Encoded values	Natural values	
	PRN (µg ml ⁻¹)	BAC (µg ml ⁻¹)
[–1,–1]	100	50
[–1,1]	100	2,000
[1,–1]	1,000	50
[1,1]	1,000	2,000
[0,0]	550	1,025
[0,0]	550	1,025
[0,0]	550	1,025
[0,0]	550	1,025

Table 1, dissolving the stock solution in sterile distilled water according to the concentration needed, and kept at 4°C until use.

Neutralising solution was prepared with following composition, per litre: 10 ml of a 34 g l⁻¹ KH₂PO₄ solution adjusted to pH = 7.2 with NaOH_(aq), 3 g soy lecithin, 5 g Na₂S₂O₃, 1 g L-histidine, 30 ml Tween 80 and deionised water (Rodríguez-López et al. 2017). This solution was sterilised by autoclaving at 121°C for 20 min and kept at 4°C until use.

Experimental design

A first order factorial design (Box 1952; Box et al. 2008) with four combinations of variables and four replicates in the centre of the domain was carried out. This kind of experimental design provides empirical information about factors without increasing the size of the assay. In addition, it is able to quantify individual and additive effects in a given experimental ambit. The natural values of the independent variables (concentration of PRN and concentration of BAC) were encoded as detailed in Table 1. Maximum and minimum values of each variable were determined in previous experiments (data not shown).

After PBS washing, PRN-BAC combinations (Table 1) were sequentially applied on the 168 h A1–A14 samples. Briefly, 1 ml of each enzymatic solution was applied for a contact time of 1 h at room temperature, statically. Next, 1.5 ml of the corresponding BAC concentration were allowed to remain for 10 min at room temperature. Finally, treated coupons were transferred to new wells containing 1 ml of neutralising solution and immersed for 30 s, which was considered the minimum time necessary for proper neutralisation according to a previous assay (data not shown). Untreated biofilm samples were used as controls. Finally, quantification of AVC, RVC and the OA by undamaged cells was carried as described above.

For better understanding, throughout this article the concentrations of the antimicrobial solutions will be displayed as [PRN, BAC] and always expressed in µg ml⁻¹.

Statistical analysis

For the factorial design, the least-squares method (quasi-Newton) was used for model fit to experimental data. The significance of the coefficients obtained in the empirical equation was determined by a Student's *t*-test ($\alpha = 0.05$). A Fisher test ($\alpha = 0.05$) was employed to test the consistency of the models.

In POA, AVC and RVC determinations, a one-way ANOVA with a *post hoc* Bonferroni test was used. Significance was expressed at the 95% confidence level ($\alpha = 0.05$) or greater.

Results

Dynamics of dual-species *L. monocytogenes* biofilm formation

Quantitative characterisation of the *L. monocytogenes* mixed biofilms was carried out by combining the outputs of the OA and plate count assays. First, the reproducibility and repeatability of both methods was assessed (Supplemental material). In addition to this, the precision of both methods in biofilm formation assays was checked by comparing the coefficients of variation of both methods at each sampling time (Supplemental material).

In samples A1–A14, the AVC values showed an increasing tendency up to a peak at 72 h of $1.21 \times 10^8 \pm 3.20 \times 10^7$ CFU cm⁻² followed by a decrease at the last two sampling times, reaching a minimum of $\sim 1.91 \times 10^6 \pm 8.44 \times 10^5$ CFU cm⁻² at 168 h (Figure 1, Table 2). The POA dynamics also showed some fluctuations, although not as sharp as in the case of AVC (Figure 1). In addition, the 72 h peak observed in AVC was not present in POA quantification (Figure 1).

Analysis of the microscopic mosaics showed a clear predominance of undamaged (green-emitting) cells over the sample (Figure 2). However, green and red cells were equally distributed along the SS surface, forming a uniform surface in which cluster formation started to be evident at 48 h (Figure 2). Clusters present at 72 h showed an intense fluorescence signal suggesting that these structures were formed by superposition of cellular layers (Figure 2). From that point onwards, these cellular aggregates became denser and more packed up to 168 h.

In G1–B52 samples, significance ($p < 0.05$) corresponding to maximum AVC values was obtained after growth for 24 h ($4.36 \times 10^8 \pm 1.82 \times 10^8$ CFU cm⁻²) (Figure 1, Table 3). From that point, the AVC outcomes decreased at the following sample times until 96 h, when the minimum AVC value was obtained ($\sim 3.13 \times 10^6 \pm 8.12 \times 10^5$ CFU cm⁻²) (Figure 1, Table 3). No statistically significant differences were observed in AVC values between 48 and 168 h (Figure 1). POA values also displayed similar dynamics where a gradual decrease from 24 h (POA = 30.86%) to 72 h (POA = 11.64%) occurred (Figure 1). Microscopy images gave evidence of cluster formation where high-density groups of red-fluorescent cells were present surrounded by a network of green-fluorescent cells (Figure 2).

Effectiveness of the sequential application of PRN-BAC on the removal of 168 h *L. monocytogenes*–*E. coli* biofilms grown on SS

Different combinations of PRN-BAC were sequentially applied on 168 h-old biofilms following a first order

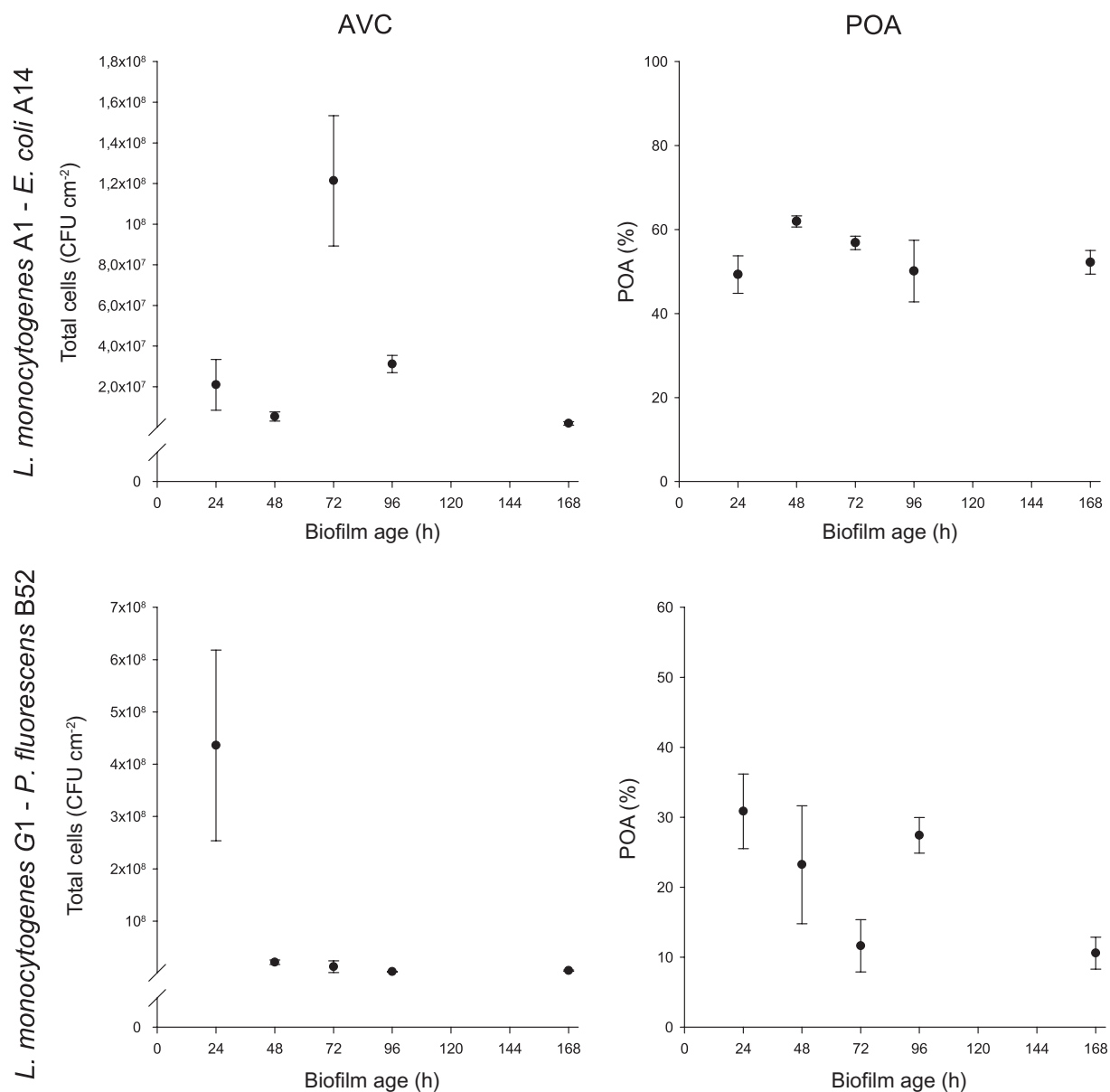


Figure 1. Representation of the total number of attached viable cultivable cells (AVC) and the percentage occupied area (POA) values of fish and dairy industry consortia obtained in biofilm formation kinetics. Error bars represent SD values ($n = 3$, for each assay).

Table 2. AVC mean values ($n = 3$) in CFU cm⁻² and standard deviations (SD) corresponding to the growth kinetics of *L. monocytogenes* A1–*E. coli* A14 dual-species biofilms on stainless steel.

Age (h)	<i>L. monocytogenes</i> A1		<i>E. coli</i> A14		Total cells	
	Mean	SD	Mean	SD	Mean	SD
24	7.07×10^6	5.88×10^6	1.38×10^7	7.08×10^6	2.09×10^7	1.25×10^7
48	1.19×10^6	3.24×10^5	4.14×10^6	2.29×10^6	5.32×10^6	2.25×10^6
72	8.33×10^5	2.63×10^5	1.21×10^8	3.21×10^7	1.21×10^8	3.20×10^7
96	1.50×10^6	1.70×10^5	2.96×10^7	2.82×10^6	3.11×10^7	2.99×10^6
168	5.13×10^5	2.32×10^5	1.40×10^6	6.84×10^5	1.91×10^6	8.44×10^5

factorial design. Quantification of the effects was carried out by combining microscopy and image analysis for occupied area (OA) determination and agar plate counts to determine the number of adhered and released viable cultivable cells (AVC and RVC, respectively) after

PRN-BAC treatments. In this study, OA will be considered as the main value for obtaining the empirical model equation, whereas AVC and RVC outcomes will be used as a supplementary values to determine the actual effects of PRN-BAC solutions.

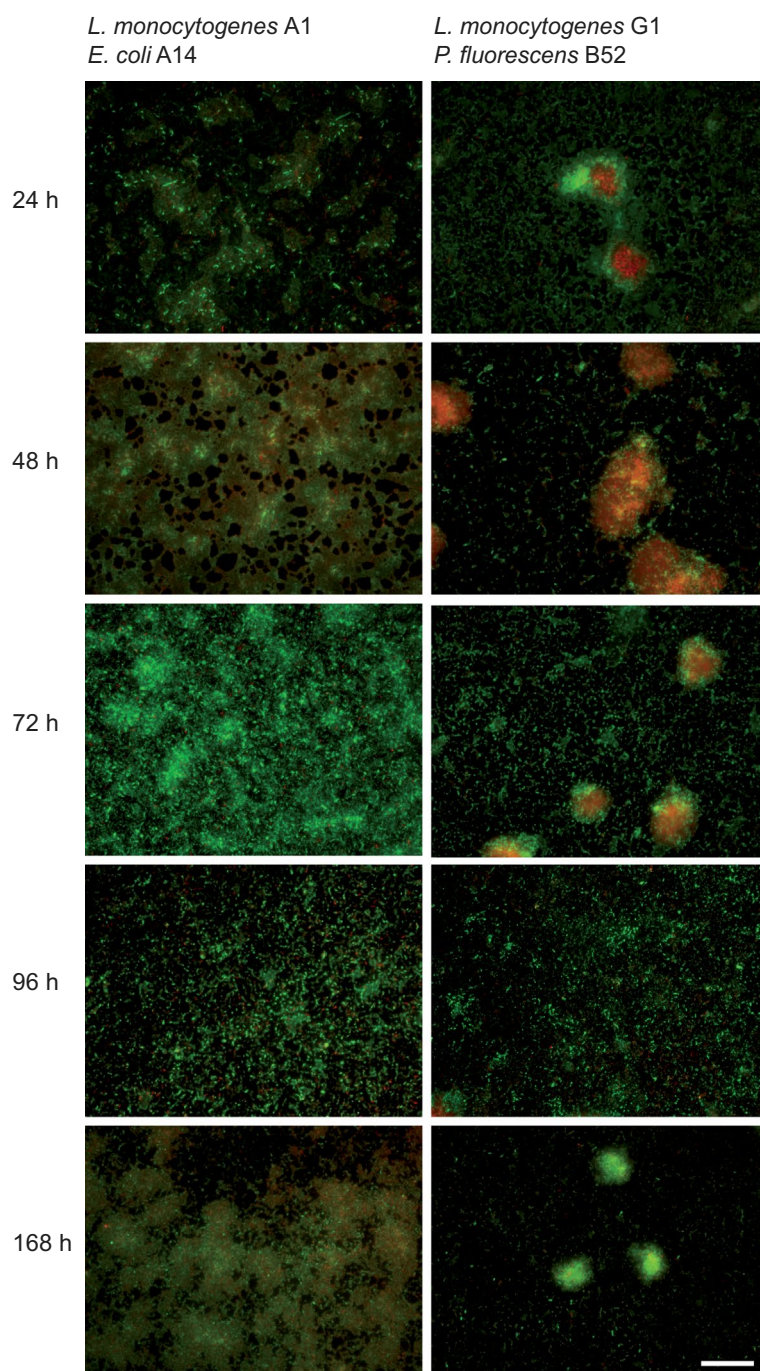


Figure 2. Formation kinetics of *L. monocytogenes* A1–*E. coli* A14 dual-species biofilms. Fluorescence microscope 40× field images obtained after LIVE/DEAD staining. Green cells represent undamaged (live) cells whereas red cells represent either damaged or dead cells (scale bar = 50 μm).

Occupied area (OA)

Empirical Equation 1 significantly ($r^2 = 0.927$) describes the combined effects of PRN-BAC sequential treatment on the occupied area by undamaged cells (according to LIVE/DEAD staining) in 168 h A1–A14 biofilms:

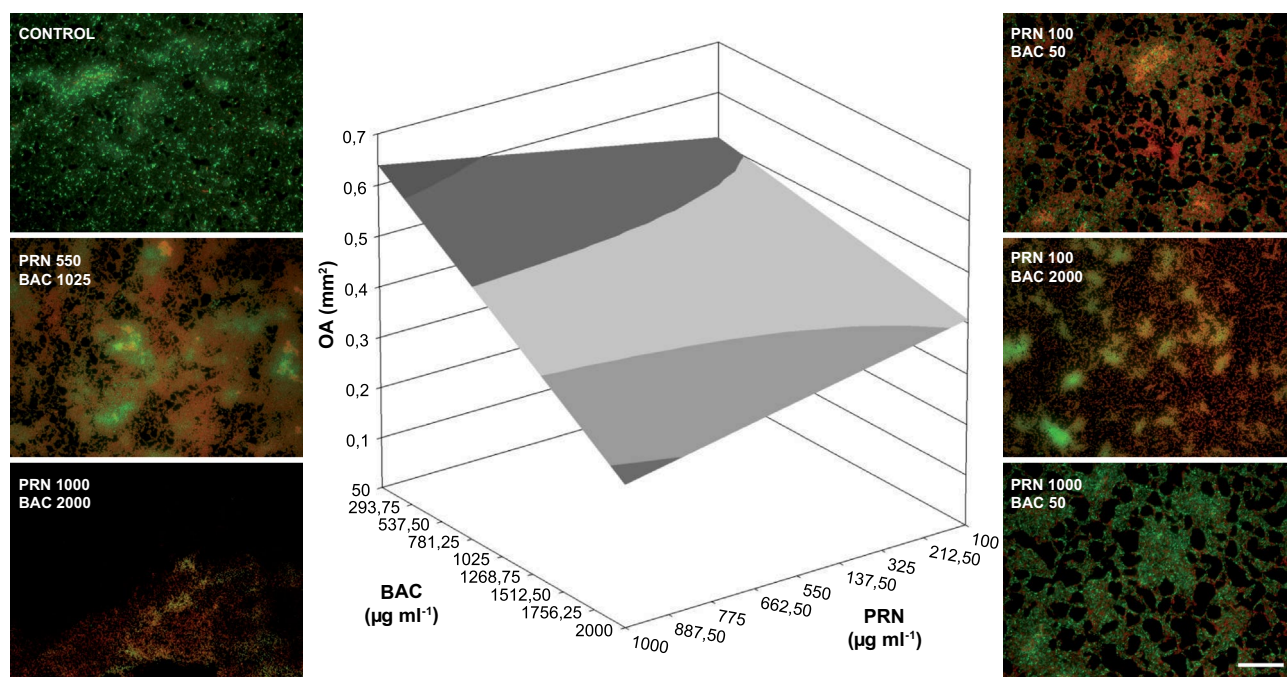
$$OA \text{ (mm}^2\text{)} = 0.46 - 0.12 \text{ BAC} - 0.06 \text{ PRNBAC} \quad (1)$$

The expected OA data according to Equation 1 after the application of PRN-BAC, together with several illustrative images are shown in Figure 3. Additionally, complete statistical data of the model can be found in Table S1.

Statistically significant coefficients in the equation indicated a negative individual effect of BAC against

Table 3. AVC mean values ($n = 3$) in CFU cm^{-2} and standard deviations (SD) corresponding to the growth kinetics of *L. monocytogenes* G1–*P. fluorescens* B52 dual-species biofilms on stainless steel.

Age (h)	<i>L. monocytogenes</i> G1		<i>P. fluorescens</i> B52		Total cells	
	Mean	SD	Mean	SD	Mean	SD
24	6.59×10^7	2.30×10^7	3.70×10^8	1.76×10^8	4.36×10^8	1.82×10^8
48	6.06×10^6	2.73×10^6	1.52×10^7	7.00×10^6	2.13×10^7	4.29×10^6
72	3.90×10^6	2.74×10^6	8.81×10^6	8.66×10^6	1.27×10^7	1.13×10^7
96	1.49×10^6	4.46×10^5	1.65×10^6	6.64×10^5	3.13×10^6	8.12×10^5
168	3.23×10^5	1.43×10^5	4.85×10^6	1.25×10^6	5.18×10^6	1.18×10^6

**Figure 3.** Central: values of occupied area (OA) expressed in mm^2 after the combined sequential application of pronase (PRN) and benzalkonium chloride (BAC) on 168 h *L. monocytogenes* A1–*E. coli* A14 biofilms. Sides: representative epifluorescence 40 $\times$ field images of control and PRN-BAC treated samples stained with LIVE/DEAD staining (scale bar = 50 μm). Numbers in each image indicate the concentration used of each component, expressed in $\mu\text{g ml}^{-1}$.

the occupied area by undamaged cells within the biofilm, thus corroborating the effectiveness of BAC as a disinfectant.

No significant effect of the application of PRN alone was demonstrated, although the negative interaction PRN-BAC proved an interactive effect of these two components. Whereas the effect of PRN increased the occupied area by undamaged cells of the biofilm at low BAC concentrations, this value was reduced as the enzyme was combined with higher BAC concentrations. Thus, the green signal (undamaged cells) was higher in samples treated at concentrations [1,000, 50] compared to those treated at [100, 50], whereas in the latter a higher red signal was observed (Figure 3). To check this effect, the experiment was repeated, yielding a similar increase in OA value (data not shown). Regardless of OA outcomes, at both experimental points an altered structure was evident compared to the control (Figure 3). Conversely, at points

[100, 2,000] and [1,000, 2,000] a higher proportion of red (damaged/dead) cells was observed, produced by higher BAC concentrations if compared to the aforementioned points. In the latter, large voids with absence of cells were also present, pointing out a substantial removal of the biofilm caused by the treatment.

The lowest expected value of OA according to Equation 1 (46% in respect of that obtained in the absence of treatments) was obtained when PRN and BAC were applied at the highest concentrations. Moreover, at point [1,000, 2,000] the majority of the remaining cells emitted a red fluorescence indicating that these cells were either damaged or dead (Figure 3).

Adhered viable cultivable cells (AVC)

No *L. monocytogenes* A1 AVC from seven-day A1–A14 biofilms were recovered from the coupons after the application of PRN-BAC combinations corresponding

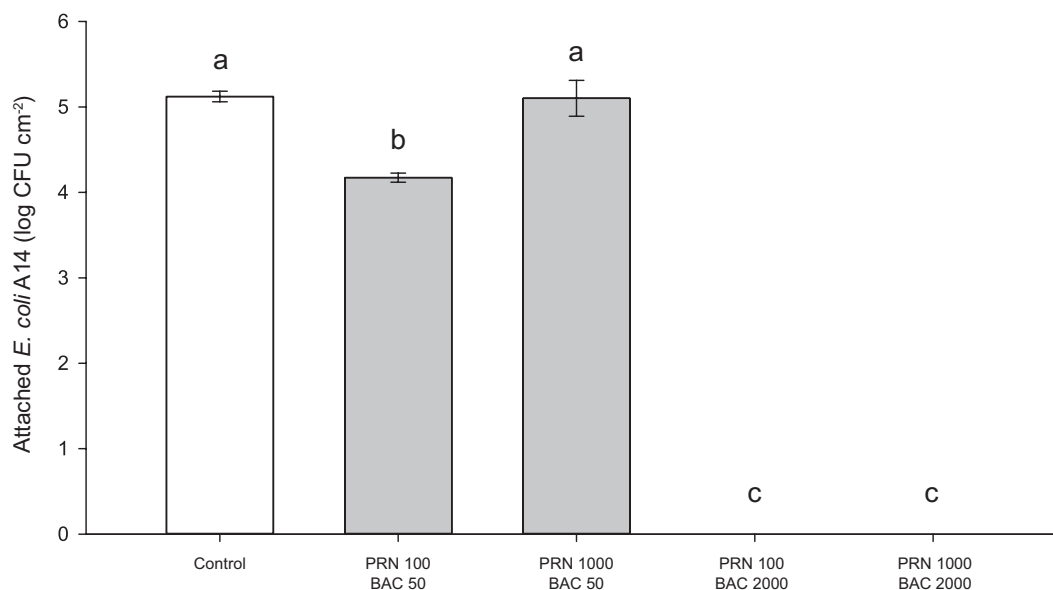


Figure 4. Remaining viable *E. coli* A14 attached cells after the sequential application of different solutions of pronase (PRN) followed by a dose of benzalkonium chloride (BAC) on 168 h *L. monocytogenes* A1–*E. coli* A14 biofilms. Values on the x-axis indicate the concentrations used of each component, expressed in $\mu\text{g ml}^{-1}$. Error bars represent SD values ($n = 3$). Different letters indicate statistical significance (one-way ANOVA, $\alpha = 0.05$).

to concentrations [100, 50], [100, 2,000], [1,000, 50] and [1,000, 2,000] $\mu\text{g ml}^{-1}$ (Figure 4). Conversely, remaining adhered cells of *E. coli* A14 were detected at the experimental points where BAC concentrations were low (Figure 4). Statistical analysis demonstrated that adhered cells of the control samples ($5.12 \pm 0.06 \log \text{CFU cm}^{-2}$) presented significant differences with experimental point [100, 50] ($4.17 \pm 0.05 \log \text{CFU cm}^{-2}$). At high PRN but low BAC concentrations, ie [1,000, 50], a higher number of *E. coli* AVC cells remained attached to the coupon ($5.10 \pm 0.21 \log \text{CFU cm}^{-2}$). Nonetheless, this value was not significantly different compared to the control samples but it was to point [100, 50] (Figure 4). This, together with the outcomes obtained in OA, suggested that at low BAC but high PRN concentrations the quantity of biofilm on the coupon increased compared with the other points of the experimental plan (Figures 3 and 4). At points [100, 2,000] and [1,000, 2,000], *E. coli* A14 AVC counts were below the level of detection, thus indicating that the elevated BAC concentrations affected the viability of the remaining attached cells (Figure 4).

Released viable cultivable cells (RVC)

No A1 strain RVC from the biofilm were recovered into the neutralising solution after the application of the PRN-BAC treatments as they were similar to the adhered cell values. Contrarily, A14 strain RVC were detected in the treatments with a low concentration of BAC, ie [100, 50] and [1,000, 50], giving significantly different values of 4.90 ± 0.19 and $5.29 \pm 0.10 \log \text{CFU ml}^{-1}$ at low and high PRN concentrations, respectively (Fig. 5), thus indicating

that PRN significantly increased *E. coli* cell detachment from the biofilm.

Discussion

Enzymes are considered good disrupters of biofilm matrices (Meireles et al. 2016). Their combined application with biocides will permit reduction in the doses of disinfectants applied and decrease environmental pollution. Furthermore, characterisation of the mixed *L. monocytogenes* biofilms potentially present on the surfaces of processing plants would permit better adjustment of the cleaning and disinfection protocols applied, and consequently avoid or decrease the generation of resistance.

Quantitative characterisation of two mixed *L. monocytogenes* biofilms was carried out in terms of the number of viable adhered cells and 2-D microscopic analysis. 2-D areal parameters have been considered as good biofilm descriptors, giving biologically meaningful information (Yang et al. 2000; Christensen et al. 2001; Jackson et al. 2001; Beyenal et al. 2004). In the present results, fluorescence microscope analysis provided information regarding 2-D structural features not detected by plate count. However, the precision of the results depends on the uniformity of the structure, giving higher values of coefficient of variation in heterogeneous biofilms (Supplemental material). This was of special relevance in G1–B52 samples in which a clear tendency to form microcolonies was observed (Figure 2). Microscopic images gave evidence of how these structures were formed by dense groups of red (damaged/dead) cells surrounded by

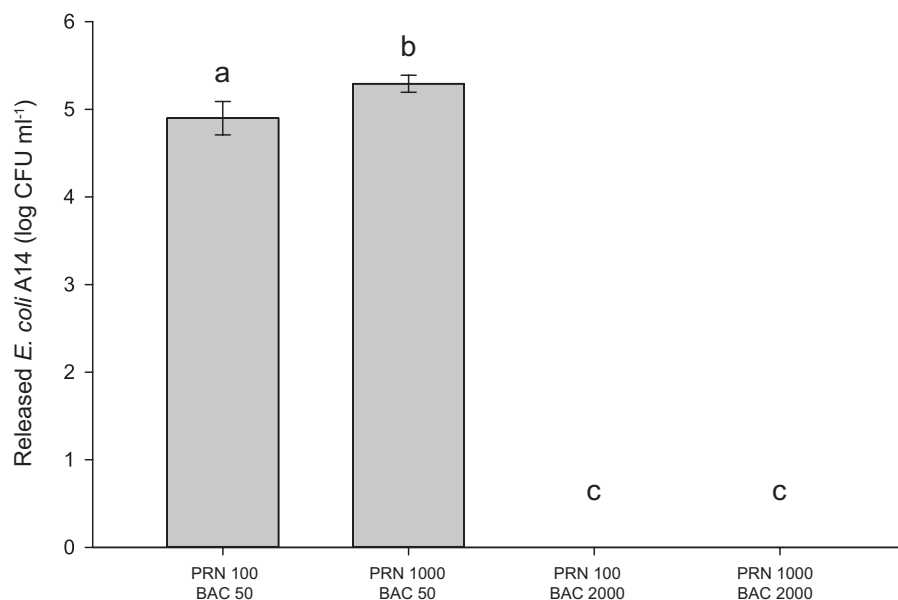


Figure 5. Viable *E. coli* A14 cells recovered from the neutralising solution after sequential application of pronase (PRN) followed by a benzalkonium chloride (BAC) dose on 168 h *L. monocytogenes* A1–*E. coli* A14 biofilms. Values on the x-axis indicate the concentrations used of each component, expressed in $\mu\text{g ml}^{-1}$. Error bars represent SD values ($n = 3$). Different letters indicate statistical significance (one-way ANOVA, $\alpha = 0.05$).

a network of undamaged (green) cells (Figure 2). Bayles (2007) pointed out the importance of dead cells as a biofilm support, anchoring the whole structure to the surface and thus improving its stability.

The effects of PRN-BAC sequential treatments were assessed on late-stage *L. monocytogenes*–*E. coli* biofilms using a first-order factorial design. Empirical Equation 1 showed no significance in the individual effects of PRN but indicated a significant individual effect of BAC and an interaction between the effects of PRN and BAC.

PRN is a mixture of various endo- and exo-peptidases (Narahashi 1970). However, although the *L. monocytogenes* biofilm matrix has a high protein content (Longhi et al. 2008; Nguyen and Burrows 2014), in all sampling times the *L. monocytogenes* A1 population was between 0.35 and 2.16 log CFU cm^{-2} lower than *E. coli* A14 (Table 2), thus the contribution to the final matrix composition of the A1 strain may be significantly lower if compared with the A14 strain. Hence, the absence of an individual PRN effect could be related to the presence of soluble protective polysaccharides in the matrix secreted by *E. coli*, becoming richer in sugar residues (Sutherland 2001). In addition, *L. monocytogenes* could have promoted this sugar-rich environment by secreting soluble polysaccharides such as teichoic acids equal to those present in the cell membrane (Brauge et al. 2016).

In order to mimic as far as possible the environmental conditions found in industrial premises, treatments were applied at room temperature, lower than optimal (Kumar et al. 2004), that may have produced a lower activity of the

enzyme. Optimal temperatures have been used in previous biofilm-removal studies involving PRN (Inoue et al. 2003; Rodríguez-López et al. 2017), as well as for other protein hydrolases (Longhi et al. 2008; Nguyen and Burrows 2014). Despite this, Orgaz et al. (2007) demonstrated the effectiveness of PRN at 25°C against *P. fluorescens* biofilms. However, the concentration of PRN used was about four times more than the maximum used in the present work. Nevertheless, microscopic images showed that, compared with the controls, biofilm structure was altered by PRN-BAC in all cases independently of the BAC concentration used (Figure 3).

At low BAC concentrations, PRN pre-treatment increased the OA occupied by the mixed biofilm ($\text{OA}_{[100, 50]} = 0.51 \text{ mm}^2$; $\text{OA}_{[1,000, 50]} = 0.64 \text{ mm}^2$) (Figure 3). Regarding the viable-and-cultivable cell quantification, an increase in the number of AVC and RVC of *E. coli* with increasing PRN was detected, whereas in *L. monocytogenes* neither AVC nor RVC were detected (Figures 4 and 5). This could be due to an intrinsic higher resistance to QACs of *E. coli*, as reported for Gram-negative bacteria (McDonnell and Russell 1999; Augustin et al. 2004). Nonetheless, as discussed later, since *L. monocytogenes* can enter into a dormant state where cells still remain alive despite the lack of growth on agar plates, its presence cannot be neglected or its contribution to the OA increase.

The unexpected increase in OA and the number of AVC of *E. coli* by PRN in the presence of low BAC concentrations can be explained by the dispersant effect of the enzyme (Figures 3 and 4). Briefly, the enzyme could

have provoked cell disaggregation in the biofilm, and the released cells could have subsequently re-adhered during the time of exposition. This hypothesis would explain, on the one hand, the observed increase in the occupied area by the re-adherence and, on the other hand, the observed increase in the number of AVC and RVC.

BAC interacts with cell membranes, promoting disruption of their integrity and cellular content leakage (Buffet-Bataillon et al. 2012; Tezel and Pavlostathis 2015). At high BAC concentrations, the results denoted the interactive effects of PRN-BAC. Indeed, minimum values of OA by undamaged cells of the biofilm were obtained at experimental point [1,000, 2,000] of the factorial design, achieving a 54% reduction in the OA by the biofilm compared to the control samples (Figure 3). These results are in agreement with the opinion of Meireles et al. (2016), who stated that a combination of enzymes and biocidal agents is desirable to obtain good biofilm biomass removal.

No AVC or RVC of *L. monocytogenes* or *E. coli* were detected in those experimental points with the highest BAC concentrations (Figures 4 and 5). However, values of occupied area indicated the presence of undamaged cells on the coupon (Figure 3). Two main reasons can explain the observed discrepancy: firstly, the lower limit of detection of the microscopic method (1 cell field⁻¹) compared with the plate count method (1.70 log CFU cm⁻²); and secondly, the presence of viable but non-cultivable cells (VBNC). Considering that the present experimental system consisted of 168 h biofilms that had been exposed to PRN-BAC treatments, it should be expected that in those biofilms the pool of VBNC cells would be significant. In this state cells do not grow *in vitro* and microscopy assays are the only alternative to detect these cells (Gião and Keevil 2014).

Moreover, it is becoming clear among microbiologists that microbial pathogens survive environmental stresses by entering into the VBNC state (Oliver 2005; Gião and Keevil 2014). This status is reversible under appropriate stimuli, so undetected pathogens can resuscitate from this dormant state, thus entailing several public health concerns (Li et al. 2014). In *L. monocytogenes* this process is multifactorial (Besnard et al. 2002). Indeed, a study carried out on biofilms grown in tap water showed that in *L. monocytogenes* the VBNC state depends not only on nutrient availability but also on the temperature (Gião and Keevil 2014).

Cell dispersion is an intrinsic process in the life-cycle of any biofilm (Petrova and Sauer 2016). Nevertheless, antimicrobial treatments can promote high dispersion after its application (Rodríguez-López et al. 2017). This has been considered as a topic of concern since it can facilitate the

dissemination of pathogens into the environment, thus becoming a feasible cause of contamination (Meireles et al. 2016; Shen et al. 2016).

The use of the combined protease-based treatment presented in this work could be useful not only in industrial settings but also in different clinical-related environments (Loiselle and Anderson 2003; Thallinger et al. 2013) to avoid biofilm formation and to remove biomass residues, thus preventing secondary colonisers further adhering after antimicrobial treatment (Cordeiro and Werner 2011). Moreover, enzymatic-based treatments have been used to sensitise biofilm so as to adjust the dose of antibiotics (Selan et al. 1993) and antiseptics (Kaplan 2009) needed for treatment of biofilm-colonised medical devices. A recent study conducted by Stiefel et al. (2016) demonstrated that the use of different species-specific enzyme mixtures increased the efficacy of commercially available cleaners in removing biofilms of *Staphylococcus aureus* and *Pseudomonas aeruginosa* potentially present in endoscopes. In addition, once bacteria are removed from the biofilm, pathogenicity factors may also be affected by enzymatic treatments. As an example, Longhi et al. (2008) observed how the infectiveness of planktonic *L. monocytogenes* in Caco-2 cells was significantly reduced after treatment for 24 h with 200 U ml⁻¹ of serratiopeptidase. This phenomenon demonstrated that despite live cells being dispersed after enzyme contact, they may represent a lower health threat.

Therefore, the main conclusion of this investigation is that it has been empirically demonstrated that the additive effects of combined treatments with PRN-BAC effectively remove late-stage *L. monocytogenes*–*E. coli* dual-species biofilms grown on stainless steel. This approach was performed in a straightforward manner, with pre-treatment with PRN combined with sequential application of a single dose of BAC. In addition, at high BAC concentrations the quantity of viable cells released from the biofilm was also significantly diminished, thus avoiding potential pathogen spread into the environment.

List of abbreviations

AVC	Adhered viable cultivable cells
BAC	Benzalkonium chloride
OA	Occupied area
POA	Percentage of occupied area
PRN	Pronase
RVC	Released viable cultivable cells
A1	<i>Listeria monocytogenes</i> A1
G1	<i>Listeria monocytogenes</i> G1
A14	<i>Escherichia coli</i> A14
B52	<i>Pseudomonas fluorescens</i> B52

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